

al-Battani / Albategnius

Opus Astronomicum

Phase 2 Draft: Arabic Transcription, English Translation, Chinese Translation

Nallino Arabic Text Volume, Pars Tertia

كتاب الزيج الصابي في حساب النجوم

《巴塔尼天文大成》第二阶段校录与翻译草稿

Source witness: Nallino Arabic-text scan, *Textum Arabicum Continens*.

This draft covers: title matter, contents opening, one prose sample, and a fixed-star table model.

Not included: no raw scan PDF, no OCR dump, no scan-wrapper replacement.

1 Source identity and reading order

The source volume is the Arabic text volume of Carlo Alfonso Nallino's edition of al-Battani: *Al-Battani sive Albatenii Opus Astronomicum, Pars Tertia, Textum Arabicum Continens*. The physical scan order is not the logical Arabic reading order. The working order used here starts from the Arabic title/opening matter and treats the early scan pages as late tables/end matter.

Latin title page	PDF page 5.
Corrigenda	PDF page 7.
Arabic title/opening	PDF pages 288 and 286–282 in reverse scan order.
Main Arabic body	PDF pages 281–61 in reverse scan order.
Tables/end matter	PDF pages 60–8 in reverse scan order.

This is a production draft, not a completed critical edition. The point of this phase is to establish a usable TeX layout and a traceable page-anchor scheme.

2 Latin title page: normalized transcription and translation

2.1 Printed Latin title

PUBLICATIONES
REALIS OBSERVATORII DE BRERA IN MILANO
No. XL, Pars III

AL-BATTANI SIVE ALBATENII
Opus Astronomicum

Ad fidem codicis Escorialensis Arabice editum,
Latine versum, adnotationibus instructum,
a Carolo Alphonso Nallino.

Pars Tertia — Textum Arabicum Continens.
Mediolani Insubrum, apud Ulrichum Hoeplium, 1899.

2.2 English

Al-Battani, also known in Latin as Albategnius, *Opus Astronomicum*: edited in Arabic according to the Escorial manuscript, translated into Latin, and furnished with annotations by Carlo Alfonso Nallino. Part III contains the Arabic text.

2.3 Chinese

巴塔尼（拉丁名 Albategnius）《天文大成》。此书由 Carlo Alfonso Nallino 依据埃斯科里亚尔写本校订阿拉伯文，配以拉丁译文和注释。本第三册为阿拉伯文本。

3 Arabic title and opening attribution

3.1 Arabic transcription, normalized

كتاب الزيج الصابئ في حساب النجوم
تأليف أبي عبد الله محمد بن جابر بن سنان الحراني الصابئ المعروف بالبتاني.

3.2 English translation

The Sabian Astronomical Tables, concerning the computation of the stars, composed by Abū ‘Abd Allāh Muḥammad ibn Jābir ibn Sinān al-Ḥarrānī al-Ṣābi’, known as al-Battānī.

3.3 Chinese translation

《萨比教徒天文表：论星辰计算》，作者为阿布·阿卜杜拉·穆罕默德·本·贾比尔·本·西南，哈兰的萨比教徒，通称巴塔尼。

3.4 Modern Arabic paraphrase

هذا كتاب في الجداول والحسابات الفلكية المنسوبة إلى البتاني، ويعرض أبوابه ومسائله في حركة الشمس والقمر والكواكب، وفي المطالع والارتفاعات والجداول اللازمة للحساب.

4 Opening contents: first reconstructed headings

The opening page states that the work is divided into a set of chapters dealing with the circle of the sphere, chords, obliquity, ascensions, solar and lunar motions, and tables. The following is a corrected first-pass contents model from the opening OCR and page image.

Arabic heading	English working translation	Chinese working translation
في تقسيم دائرة الفلك وضرب الأجزاء بعضها في بعض	Division of the celestial circle and multiplication of arc-parts.	天球圆周的划分及弧分相乘。
في معرفة أوتار الأجزاء وأنصاف أوتارها	Chords of arcs and half-chords, with tabular use.	弧的弦与半弦及其表用。
في مقدار ميل فلك البروج عن فلك معدل النهار	The obliquity of the ecliptic relative to the equator.	黄道相对于赤道的倾角。
في معرفة مطالع البروج في الفلك المستقيم	The right ascensions of the zodiacal signs.	黄道十二宫在正球中的升度。
في أحوال الكواكب الثابتة	States and positions of the fixed stars.	恒星的位置与状态。

5 Prose sample: fixed stars

5.1 Arabic transcription

قال: وأما أحوال الكواكب الثابتة في طلوعها وغروبها وتوسطها السماء ومكثها فوق الأرض وتحتها، فقد ذكر القدماء مواضعها في الطول والعرض. ونحن نضع ههنا ما صحّ عندنا بالرصد، ونرتب أسماءها وما يتصل بها من الأجزاء والدرجات.

5.2 English translation

He says: As for the states of the fixed stars in their rising, their setting, their culmination in the sky, and the duration of their stay above and below the earth, the ancients have recorded their places in longitude and latitude. We set down here what has been established for us by observation, arranging their names and the related parts and degrees.

5.3 Chinese translation

他说：至于恒星的状态，包括其升起、落下、过中天，以及在地平线上下停留的时间，古人已经记录了它们的黄经和黄纬。我们在此列出经观测所确认者，并整理其名称以及相关的度、分。

6 Fixed-star table model: column structure

The star tables are not safe to transcribe as prose. The columns are therefore modeled as a structured table. Numerical columns are intentionally not finalized in this phase; the goal here is to stabilize the row identity and multilingual schema.

Arabic column	English gloss	Chinese gloss	Use in next pass
أسماء الكواكب الثابتة	names of the fixed stars	恒星名称	row identity
الأجزاء التي معها تطلع	parts/degrees with which they rise	同升度分	numeric column
الأجزاء التي معها تتوسط السماء	parts/degrees with which they culminate	过中天度分	numeric column
الأجزاء التي معها تغيب	parts/degrees with which they set	同没度分	numeric column
نصف مكثها فوق الأرض	half the duration above the earth	地平线上半停留 时长	numeric column
ارتفاعها في وسط السماء	altitude at culmination	中天高度	numeric column

7 Fixed-star table pilot rows, pages 8–9

Arabic row name	English working translation	Chinese working translation
الذي تحت السرير	the star under the Throne or Seat	在座下之星
الثلاثة منها	the three stars from it	其中三星
الذي من الشجاع	the one from Hydra / the Serpent	长蛇座中之一星
بطن فرس قنطورس	the belly of the Horse of Centaurus	半人马马腹之星
وسط نخذ الفرس	the middle of the horse's thigh	马股中央之星
حافر هذا الفرس	the hoof of this horse	此马蹄之星
رجل هذا الفرس	the foot or leg of this horse	此马足之星
المضيء من كواكب الإكليل	the bright one among the stars of the Crown	冠星中明亮者
أول مصب الماء من منكب الدلو	the first outflow of water from the shoulder of Aquarius	宝瓶肩出水之首星
وسط مصب الماء	the middle of the water-stream	水流中央之星
آخر مصب الماء	the last part of the water-stream	水流末端之星
كفة الميزان الجنوبية	the southern pan of the Balance	天秤南盘
كفة الميزان الشمالية	the northern pan of the Balance	天秤北盘
قلب العقرب	the heart of Scorpio	天蝎之心
عرقوب الرامي	the heel of the Archer	射手之踵
عين الرامي	the eye of the Archer	射手之眼

8 Corrigenda model

The printed corrigenda page is mixed Arabic and Latin reference notation. The safe TeX model is a three-column critical apparatus: printed page/line, printed reading, corrected reading. A sample follows.

Reference	Printed reading	Corrected reading
p. 1, l. 18	وتجزئة	وتجزئة
p. 4, l. 9	--	راجع موضع التصحيح في الأصل
p. 19, l. 19	حاجتنا	حاجتنا
p. 28, l. 15	الدائرة	الدائرة

The apparatus above is a model, not the completed corrigenda transcription. The next pass should capture every corrigendum against the page image, because the OCR is especially unstable in this section.

9 Glossary seeded from this phase

Arabic	English	Chinese
الكواكب الثابتة	fixed stars	恒星
فلك البروج	ecliptic / zodiacal sphere	黄道圈
فلك معدل النهار	celestial equator	赤道圈
مطالع البروج	ascensions of the signs	黄道十二宫升度
الأوتار	chords	弦
نصف الوتر	half-chord	半弦
الطول	longitude	经度
العرض	latitude	纬度
وسط السماء	culmination / mid-heaven	中天

10 Additional chapter-heading samples from the Arabic body

These headings are normalized from the OCR candidate list and then checked against the rendered source-page class. They are included to start the chapter-range map; numerical tables still require visual verification.

Arabic heading	English working translation	Chinese working translation
باب معرفة الظل والارتفاع في أجزاء البروج في كل بلد	On knowing shadows and altitudes in the parts of the zodiac for every locality.	论各地黄道各度的影长与高度。
باب معرفة عروض القمر	On knowing the latitude of the Moon.	论月纬的求法。
باب كسوف القمر	On the eclipse of the Moon.	论月食。
باب معرفة أوقات تحويل السنين	On knowing the times of the conversion of years.	论历年换算的时刻。
باب معرفة أوائل شهور الروم	On knowing the beginnings of the Roman months.	论罗马月份的月首。
باب معرفة مطالع البروج في الفلك المستقيم	On knowing the ascensions of the zodiac in the right sphere.	论正球中黄道诸宫的升度。

11 Translation style sample

11.1 Arabic

إذا أردت معرفة مطالع البروج في الفلك المستقيم، فخذ أوتار الأعشار المرسومة على الرسم المتقدم في صدر الكتاب، ثم انظر إلى ما يقابلها في الجداول.

11.2 English

When you wish to know the ascensions of the signs in the right sphere, take the chords of the ten-degree divisions as drawn in the figure already given at the beginning of the book, and then look to the corresponding entries in the tables.

11.3 Chinese

若要求正球中黄道诸宫的升度，则取本书前面图中所绘每十度分弧的弦，再查表中相应的条目。

12 Fixed-star table: data structure for scaling

The finished full transcription should store star-table material in CSV and generate TeX from it. The column order below is the working schema for the rows already seeded in this phase.

Field	Script	Meaning	Comment
source page	Latin	physical scan page	required for traceability
logical order	Latin	Arabic reading-order position	protects against reversed scan order
Arabic name	Arabic	row name in Nallino table	first correction target
longitude degree	Arabic numeral/text	degree component	blank until visually verified
longitude minute	Arabic numeral/text	minute component	blank until visually verified
rising with	Arabic numeral/text	ascensional entry	table-specific
culminating with	Arabic numeral/text	mid-heaven entry	table-specific
setting with	Arabic numeral/text	setting entry	table-specific
English gloss	Latin	working translation	revise against star-name dictionaries
Chinese gloss	CJK	working Chinese translation	revise after English stable

13 Phase 2 completion note

This PDF is a modern TeX draft, not a scan wrapper. The included material now gives a working multilingual architecture for the al-Battani project: title, source identity, Arabic opening attribution, logical page anchors, a table schema, a first fixed-star row list, a corrigenda model, and a seeded glossary. The next additions should be made by extending the same TeX source and CSV table, not by placing the raw scan in a replacement folder.